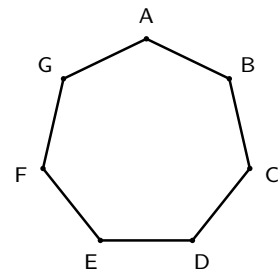
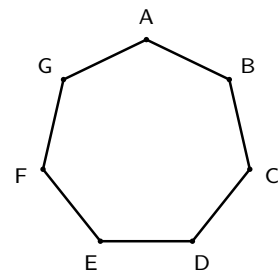
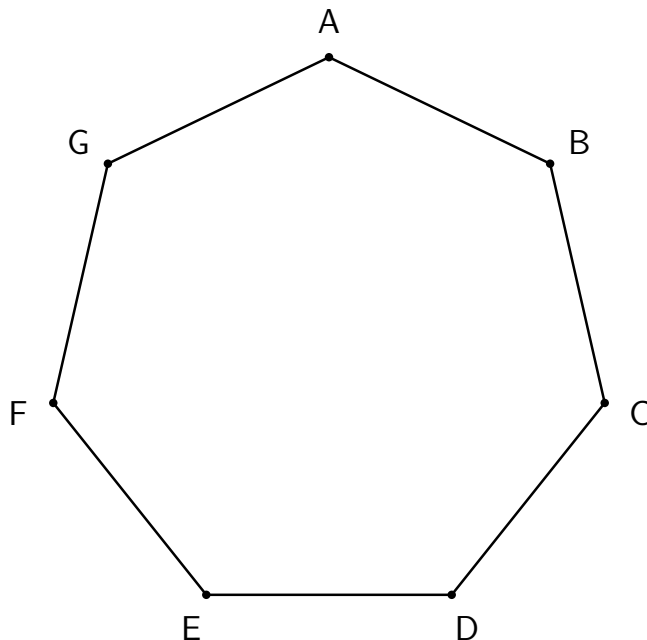
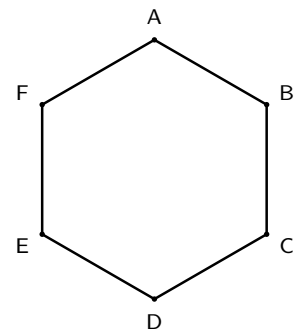
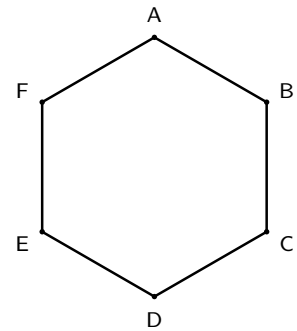
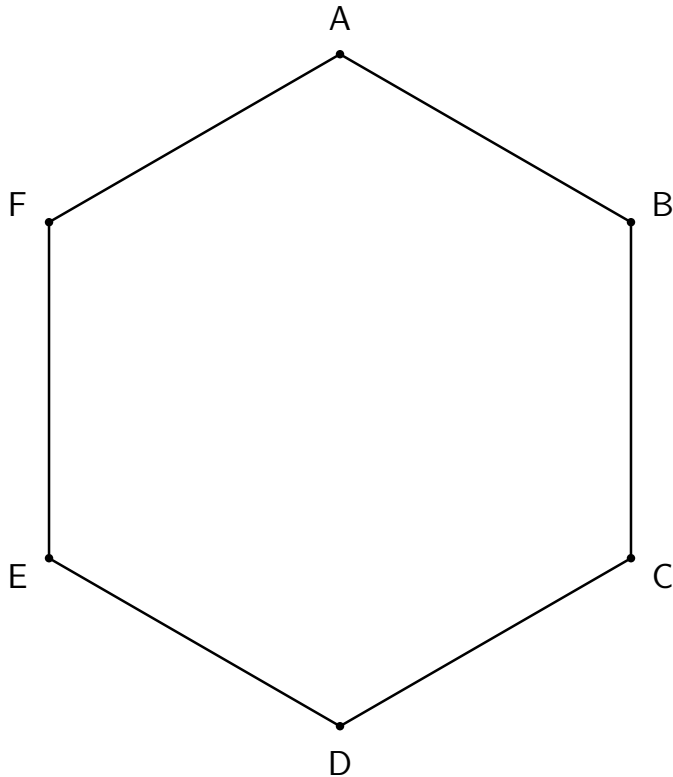
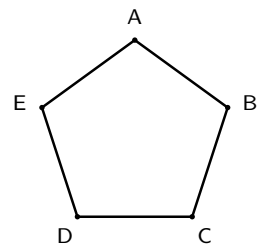
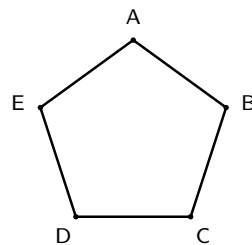
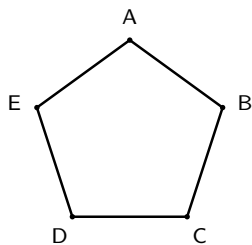
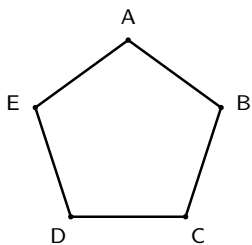
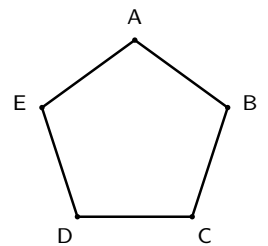
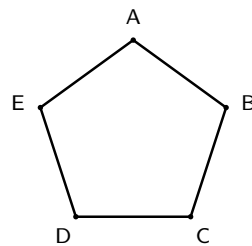
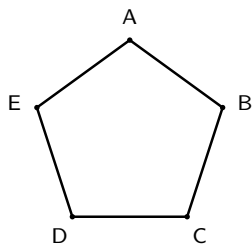
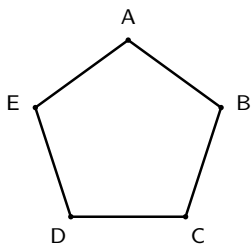
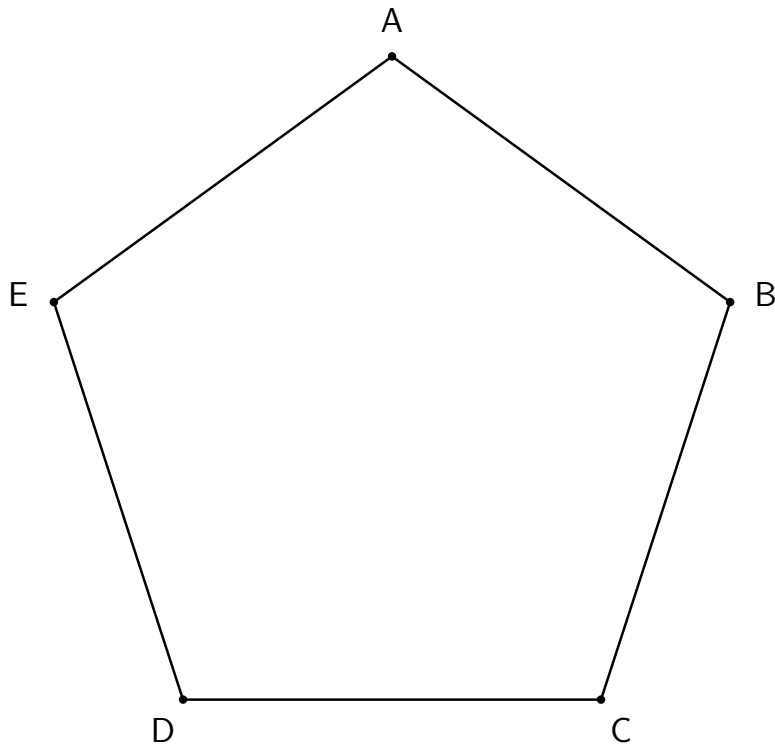


Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

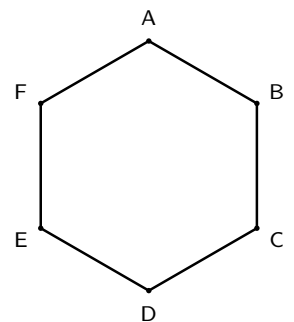
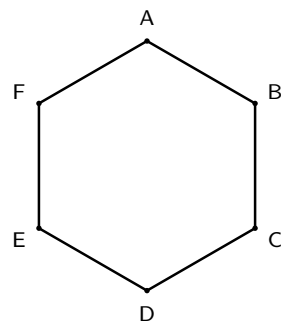
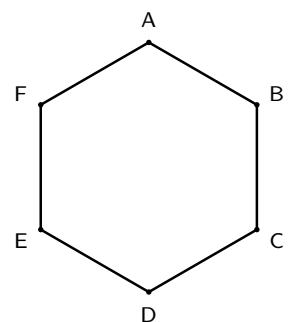
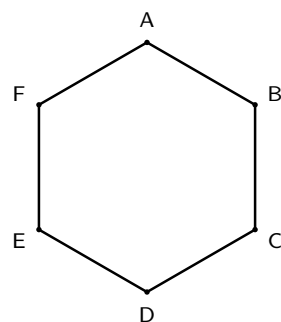
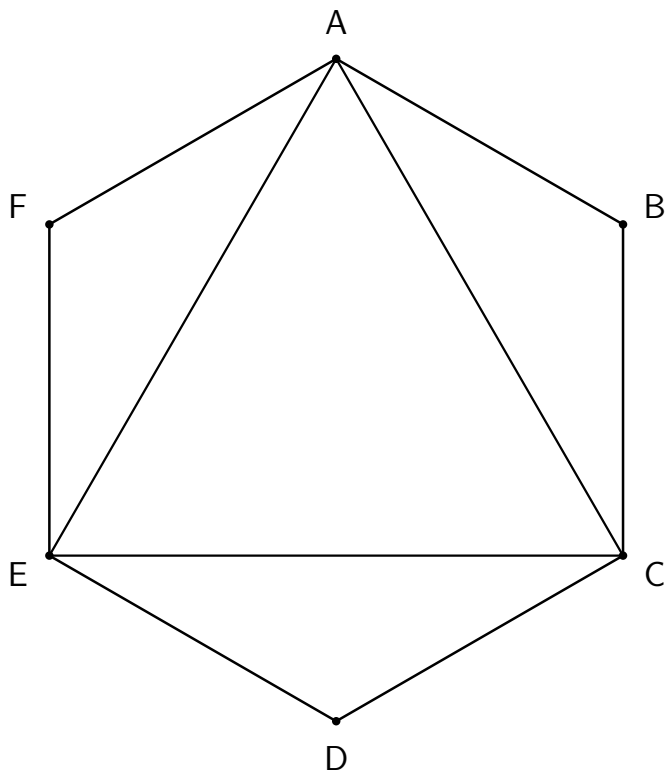
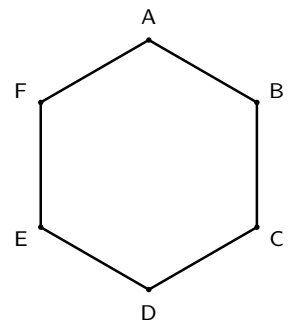
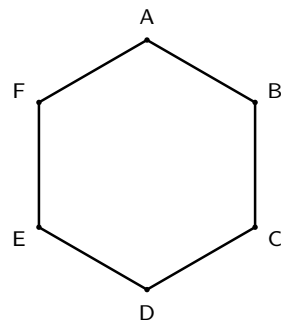
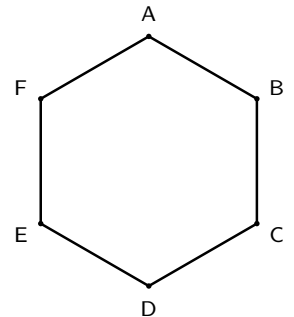
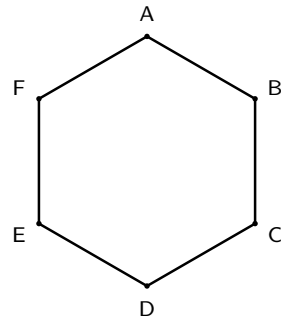
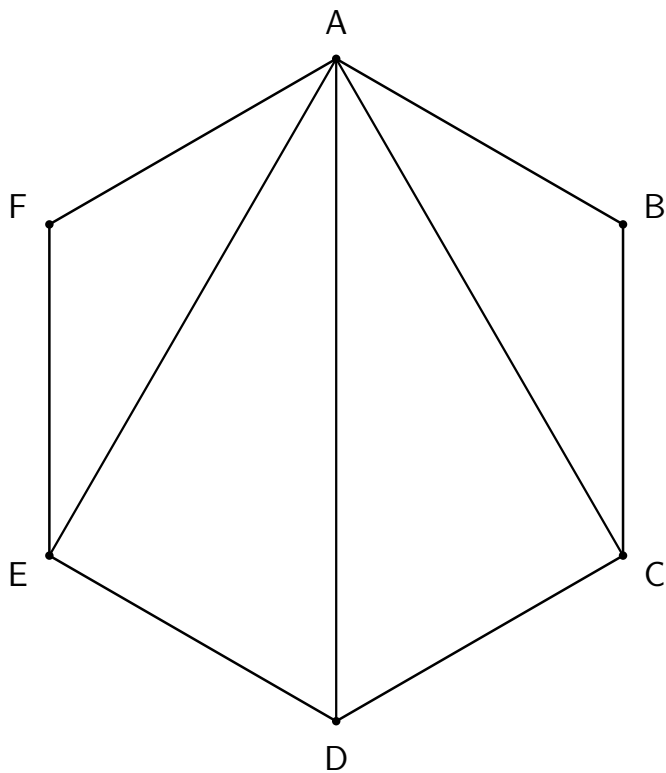
Problem 1: Fill each polygon in two different ways. Can the two fillings of one polygon have different numbers of triangles?



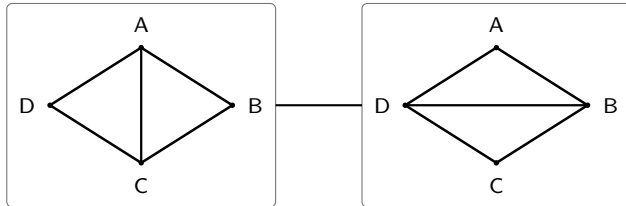
Problem 2: Find every filling of this five-corner polygon. Organize your drawings so that you can explain why none are missing.



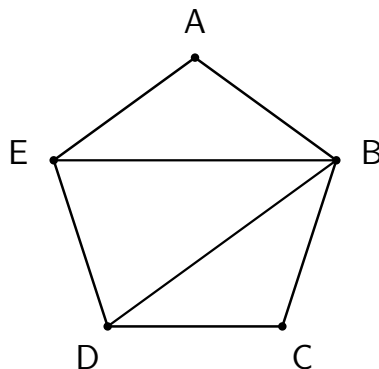
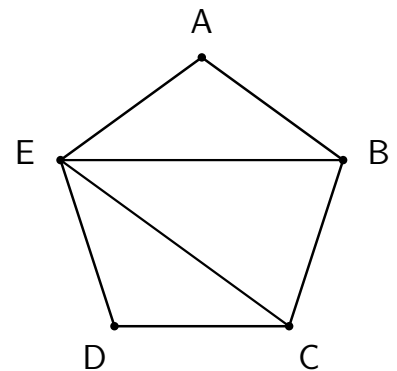
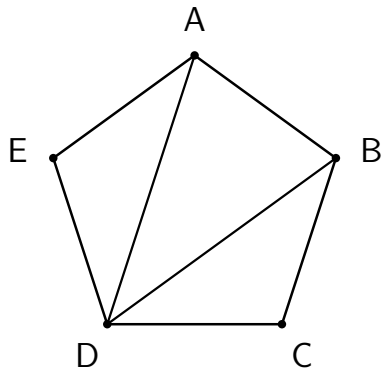
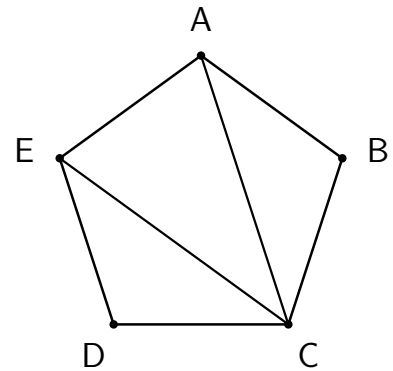
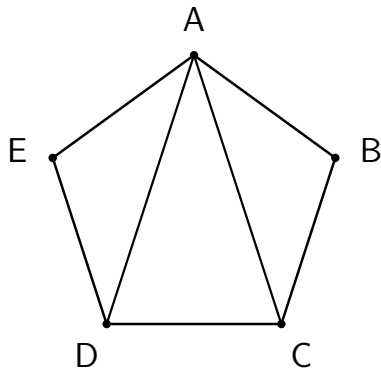
Problem 3: Erase one inside line and replace it with a different line so the polygon stays filled with triangles. Find every result of one such change from each drawing.



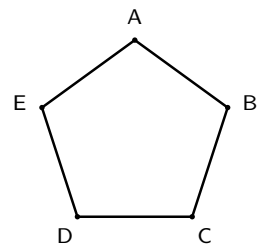
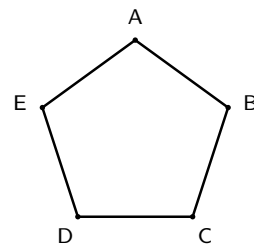
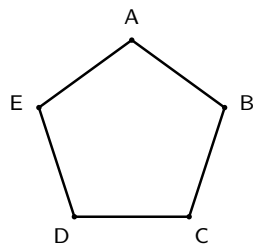
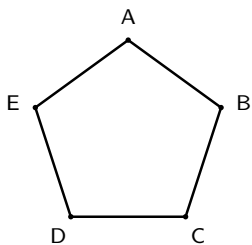
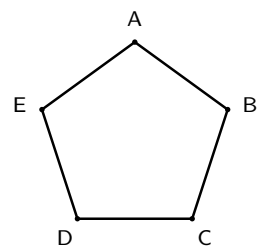
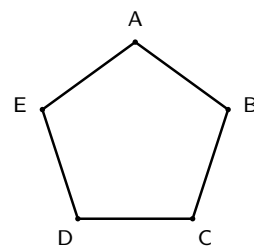
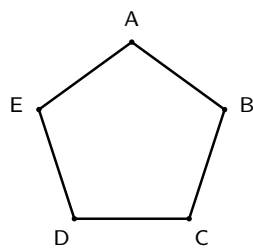
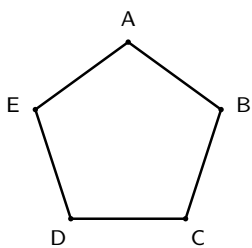
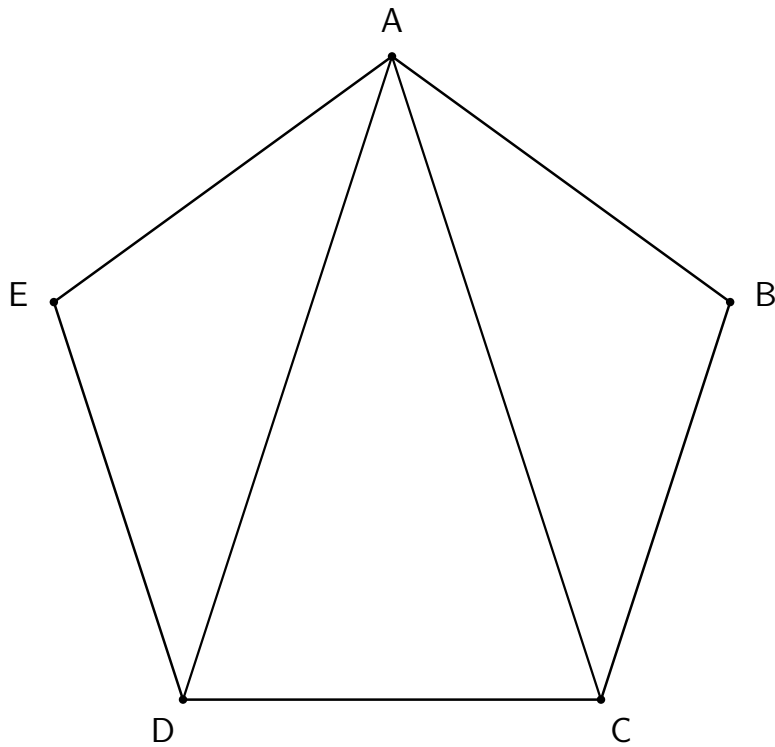
Problem 4: A change of one inside line that keeps the polygon filled with triangles is called a flip. Join two drawings when one flip changes one into the other. Include every possible join.



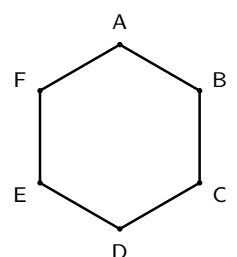
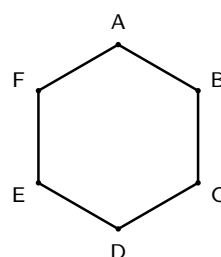
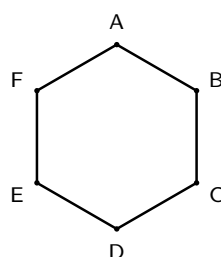
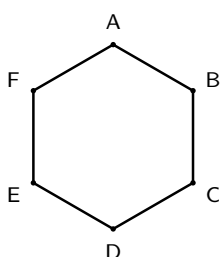
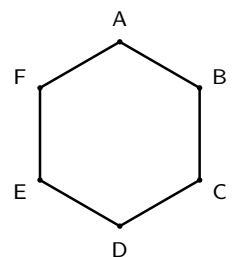
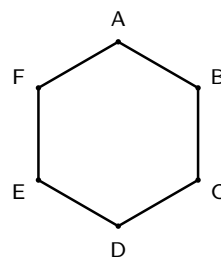
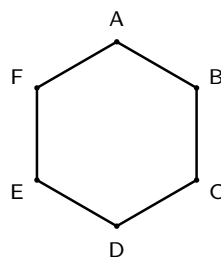
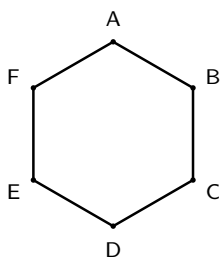
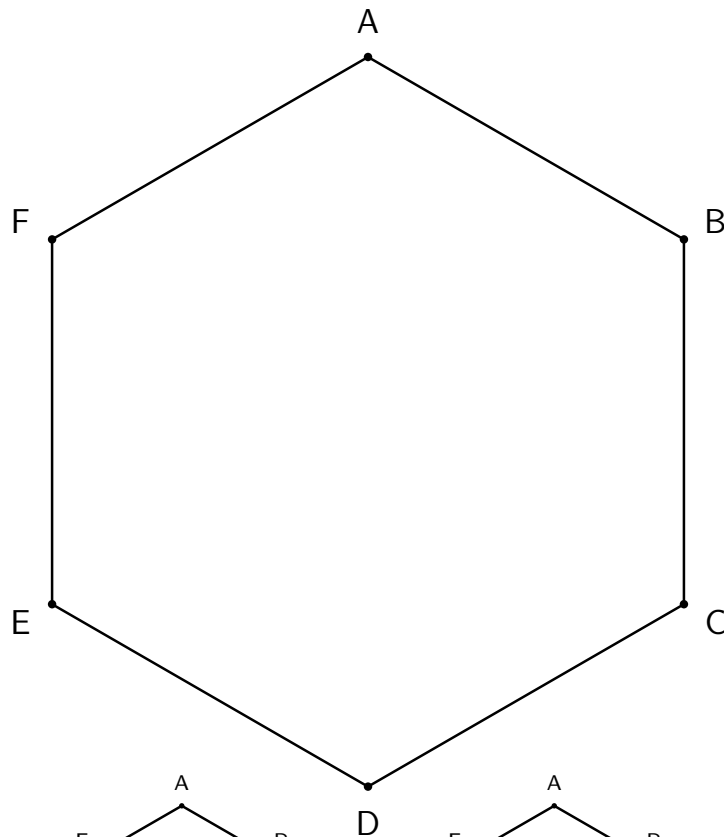
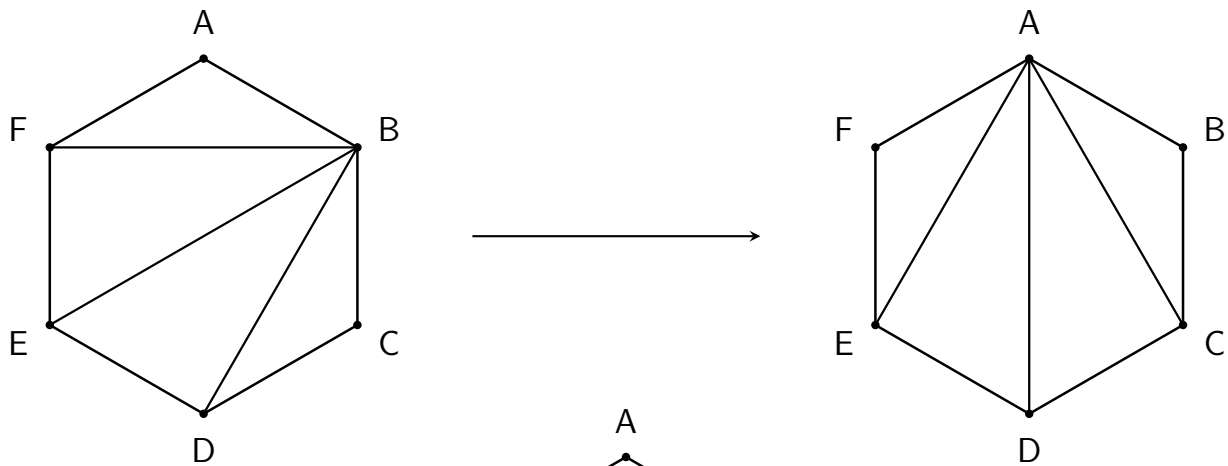
Example: erase AC, draw BD. The join connects whole fillings; it is not an inside line.



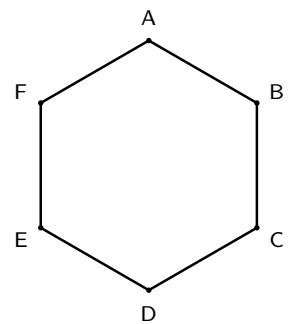
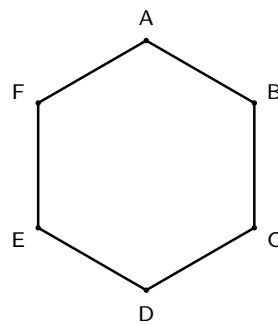
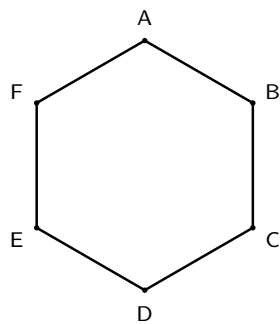
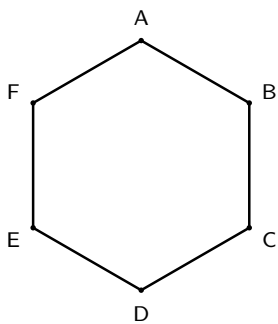
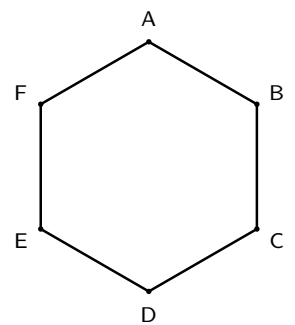
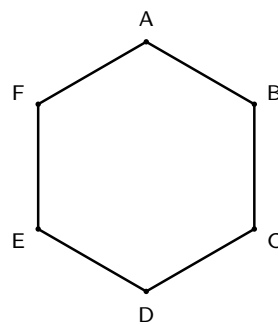
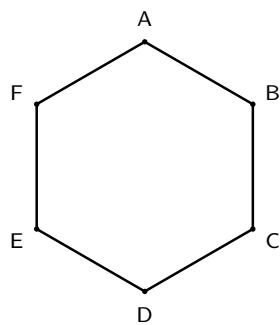
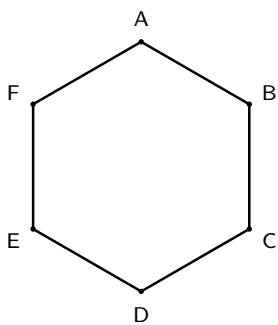
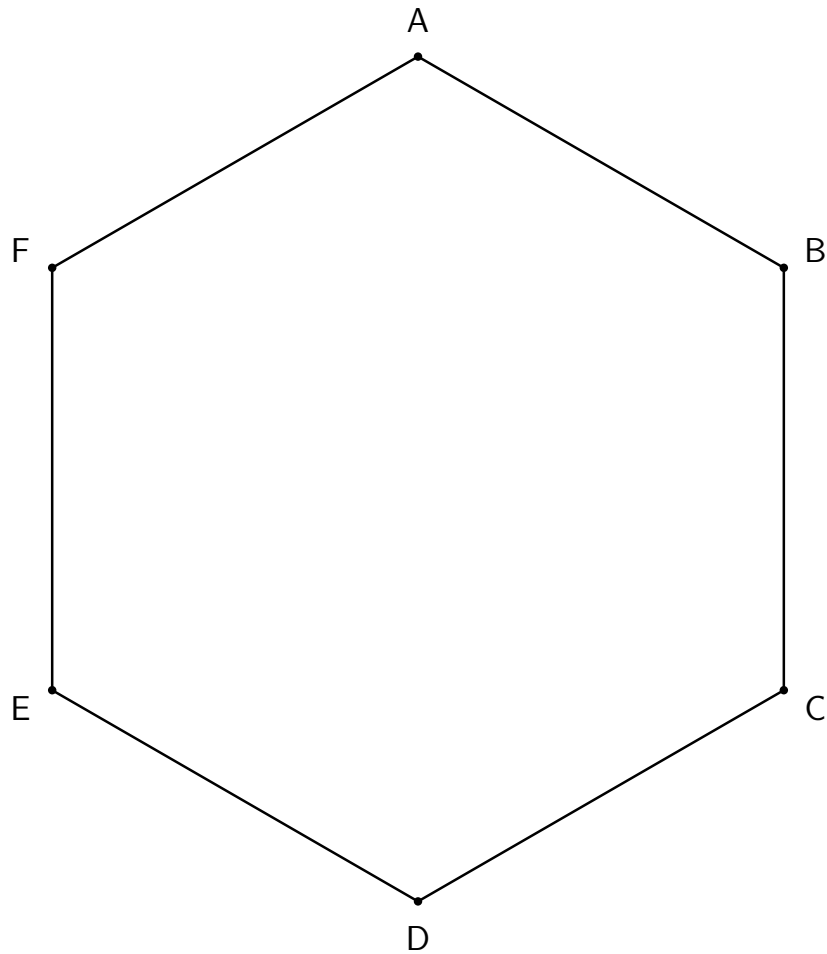
Problem 5: Start with this filling and return to it after an odd number of flips. Show every filling along your route.



Problem 6: Change the first filling into the second using as few flips as you can. Find another route with a different number of flips, without visiting any filling twice.



Problem 7: Find every filling of this six-corner polygon. Organize your drawings so you can tell whether any are missing or repeated.



Problem 7 (continued): Extra hexagons for your collection. Use these to continue Problem 7 on the previous page.

